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Professional Summary

- Accomplished Senior AI/ML Engineer and Data Scientist with 7 years of experience architecting, building, and operating production-grade ML platforms and pipelines across fintech, e-commerce, insurance, and telematics domains. Deeply skilled in end-to-end ML delivery (training → deployment → monitoring → retraining), leveraging strong Python/SQL, modern cloud platforms (AWS, Azure, GCP), and distributed data engineering stacks (Spark, Databricks, Kafka, Snowflake) to deliver reliable ML systems at scale. Hands-on expertise in MLOps best practices—CI/CD for ML, model versioning/registry, reproducibility, governance gates, rollout/rollback strategies, and observability (latency, error rate, drift, cost)—to ensure resilient and compliant production deployments.
- Demonstrated success leading cross-functional teams (data science, engineering, product, and DevOps/SRE) to deliver high-impact, business-critical AI products: demand forecasting, fraud detection, anomaly detection, semantic search, customer segmentation, churn prediction, and dynamic pricing. Adept at strengthening ML system reliability and operational excellence through standardized deployment patterns (batch + real-time inference), automated testing and release controls, and monitoring/alerting frameworks that tie model health metrics to business KPIs. Experienced partnering with risk, compliance, and security stakeholders to support audit readiness and regulatory expectations (SOC2, GDPR) in data/model workflows.
- Known for hands-on leadership, process improvement, and technical depth—owning the full lifecycle: data strategy, advanced analytics, model development, experimentation/A/B testing, CI/CD automation, cloud deployment, monitoring and drift detection, incident/runbook-driven operations, and executive reporting. Published author with peer-reviewed research in LLM-based semantic search for e-commerce; comfortable integrating LLM-based workflows where valuable, while maintaining rigorous engineering standards for production reliability. Thrive in fast-paced, global environments; passionate about enabling scalable, secure, and observable ML systems that drive measurable business value.

Technical Skills:

MLOps / ML Platform Engineering (Core):

- End-to-end ML pipelines: training → deployment → monitoring → retraining; model lifecycle management at scale
- CI/CD for ML: Jenkins, GitLab CI, Azure DevOps; unit/integration testing, reproducible builds, gated promotions (staging→prod)
- Model management & governance: MLflow (experiment tracking + model registry), versioning, approvals, rollback, shadow/champion–challenger
- Monitoring & reliability: model/data drift detection, performance telemetry, alerting, runbooks; service SLIs/SLOs (p95 latency, error rate, throughput), cost per inference/query

Serving / Infrastructure / Deployment:

- Docker, Kubernetes, Git; containerized ML workloads, autoscaling, blue/green or canary-style rollouts, safe rollback patterns
- REST/JSON APIs; batch + real-time inference patterns; production support readiness (release controls, incident triage workflows)
- Infrastructure as Code (Terraform / CloudFormation) – provisioning repeatable environments, policies, and deployments

AI/ML & Data Science:

- Deep Learning (TensorFlow, PyTorch, attention/LSTM models, XGBoost, Scikit-Learn)
- Feature Engineering, Predictive Modeling, Forecasting, Churn/Propensity Modeling, Recommendation Systems
- Real-time Analytics (Kafka, Spark Streaming, Delta Lake, Kinesis)
- Natural Language Processing (NLP), LLMs (OpenAI GPT-4, FLAN-T5, Sentence-BERT, LangChain, LangGraph, RAG, prompt engineering)

Data Engineering & Platforms:

- Data Pipelines: Airflow, dbt, ETL/ELT, Data Lake, Data Mart Design
- Cloud: AWS, Azure, GCP (S3, Glue, Redshift, Synapse, BigQuery, Data Factory)
- Databases: MySQL, SQL Server, Snowflake, Hive, Delta Lake
- Data Visualization: Power BI, Tableau, Plotly, executive dashboards

Product, Analytics, and Leadership:

- A/B Testing, Experimentation, User Segmentation, Cohort Analysis
- Business Intelligence, KPI/Metric Design, Data-driven Product Roadmaps
- Team Leadership & Mentorship (Data Scientists, ML Engineers, Analytics, DevOps), Cross-functional collaboration
- Agile/Scrum Project Management, Release Train Engineering, stakeholder management, executive reporting.

Data Governance & Compliance:

- Data Quality, Metadata Lineage, Data Contracts
- Security: PII handling, Encryption (at rest/in flight), Role-based Access Control (RBAC)
- Regulatory: SOC2, GDPR, audit readiness; compliance-aware ML delivery and approvals.

Other Tools & Tech:

- FAISS, Pinecone (Vector DBs), Alation (data catalog), Azure Monitor / App Insights (telemetry), Confluence/Jira, ServiceNow Change/CAB

Professional Experience:

Client: Fiserv Remote, US

May 2025 – Present

Senior AI/ML Engineer

- Spearheaded the AI-driven transformation of Fiserv's client servicing and merchant experience platforms: defined the north star KPIs (customer self service rate, first contact resolution, merchant onboarding time, dispute cycle lead time) and mapped each major initiative to those outcome metrics.
- Designed and implemented LangChain based agentic workflows for client servicing, dispute resolution, and cardholder inquiry use cases: modeled multi-step graphs (retrieval → reasoning → tool-calling → escalation) over Snowflake and Pinecone-backed vector stores, enabling autonomous resolution of common requests and reducing average handling time by ~18%, while enforcing production guardrails and auditability.
- Built NLP preprocessing + query understanding layers for service chat and case notes using NLTK (normalization, keyword/phrase cues, lightweight entity extraction) to improve intent routing and retrieval query quality, reducing noisy retrieval and improving first-pass resolution consistency.
- Implemented RAG grounding + citation patterns for high-risk domains (KYC, disputes, merchant onboarding) by retrieving policy/procedure context from governed sources and enforcing "answer-with-evidence" behavior; reduced hallucination-driven rework and improved compliance alignment.
- Integrated LangChain/LangGraph agents with Pinecone semantic retrieval layers and Fiserv Developer Studio SDKs/APIs and internal payment ledger services (merchant onboarding, KYC, push payments, dispute servicing), wiring tools into graph nodes so agents could safely read/write system-of-record data under strict guardrails; reduced cross-system integration defects by ~30% and eliminated large classes of manual swivel-chair work
- Built predictive analytics models for payment fraud, merchant category code (MCC) classification, and transaction anomaly detection: deployed transformer-based architectures to monitor real time ledger flows, implemented alerting logic and thresholding to reduce false positives and improve detection latency by over 25% in production monitoring loops.
- Architected end-to-end MLOps pipelines for training, fine-tuning, deployment, monitoring, and rollback across multi-tenant payments platforms: leveraged CI/CD via Azure DevOps/Git/Docker/Kubernetes, integrated MLflow for experiment tracking + model registry, and enforced gated promotions (tests, approvals, change controls) for reliable staging→prod releases.
- Established and matured a data governance framework for AI/ML: built a federated data stewardship model, partnered with the Data Intelligence team around the Alation catalogue initiative (metadata discovery, lineage,

search) to elevate AI readiness of legacy payment and ledger systems; enabled controlled natural-language access while maintaining RBAC and audit expectations.

- Partnered with business stakeholders (merchant & banking segments, risk, compliance, customer experience) to translate strategic priorities into AI product roadmaps: for example, a priority on reducing merchant onboarding friction led to a conversational agent for onboarding and KYC (know your customer) tasks, resulting in a 20% reduction in average onboarding time.
- Led a team of senior data scientists, ML engineers, data engineers and AI product owners across multiple geographies: instituted agile release trains (aligned with Azure DevOps pipelines), implemented branching strategies, gated release controls, and achieved >95% change success rate in production AI deployments.
- Designed monitoring, metrics and continuous improvement frameworks: built dashboards (e.g., Power BI, embedded in Fiserv internal MI) for tracking model performance (accuracy, latency, drift, cost per query), and operational SLIs/SLOs (p95 latency, error rate, throughput) with alerting and runbooks; applied A/B tests and champion challenger experiments achieving measurable outcomes (14% uplift in self service resolution, 8% improvement in cross sell conversion via AI recommendation engine).
- Acted as the strategic AI/ML advisor to Fiserv senior leadership and key clients: helped evaluate open source vs proprietary LLMs, set up an AI Centre of Excellence (CoE), defined ethical AI and bias mitigation policies, guided data privacy/compliance posture, and standardized production-readiness practices (release gates, rollback playbooks, incident triage workflows) for operational excellence.

Environment: Python / PyTorch / TensorFlow, LangChain, OpenAI or equivalent LLMs, Azure OpenAI or comparable cloud AI services, Snowflake, Alation (data catalogue), Azure DevOps, Git, Docker, Kubernetes, Power BI, REST/JSON APIs, payments/ledger systems, real-time event streaming, monitoring.

Client: Mercor Remote, US

Jan 2025 – May 2025

Applied Data Scientist

- Designed domain-specific evaluation worlds and datasets to stress-test Claude LLM on complex, real-world reasoning tasks, ensuring coverage across edge cases, ambiguity, multi-step constraints, and failure modes.
- Authored high-signal prompts and task suites (multi-part instructions, adversarial variations, and grounded scenarios) to measure model capabilities in planning, retrieval, structured output, and instruction-following consistency.
- Built NLP-driven error taxonomies and automated response analysis using NLTK (tokenization/normalization, pattern-based labeling) to categorize failures (hallucination, tool misuse, constraint violations), improving reviewer consistency and benchmark reliability.
- Collaborated cross-functionally with data scientists and data engineers to operationalize evaluation pipelines (dataset iteration, prompt versioning, and QA), translating model behavior insights into improved task design and measurable performance tracking.
- Created detailed scoring rubrics and grading frameworks (criteria, partial credit rules, pass/fail thresholds) to evaluate correctness, safety, and adherence; operationalized reviewer calibration to increase inter-rater agreement and scoring stability.
- Built an RL-style evaluation environment for tool-calling agents, including simulated APIs, tool schemas, and success metrics; added RAG test scenarios (retrieval + citation + grounding checks) to evaluate retrieval faithfulness and refusal behavior under realistic constraints.

University of Arizona Tucson, US

Jan 2024 – Dec 2024

Applied ML Research Assistant

- Designed and deployed a production-scale semantic search and retrieval system for conversational queries using fine-tuned Sentence-BERT embeddings, enabling high-precision product discovery across million-scale catalogs with low-latency vector search.
- Built and operated vector search infrastructure leveraging Pinecone and FAISS for scalable embedding storage, similarity search, and metadata-based filtering, supporting real-time retrieval and downstream ranking workflows.
- Implemented RAG retrieval pipelines using LangChain and LangGraph, orchestrating multi-step workflows for query understanding, intent extraction, retrieval, and constraint-based filtering to improve relevance and explainability.

- Built query parsing + normalization utilities using **NLTK** (text cleanup, token rules, synonym/phrase handling where applicable) to reduce query noise and improve downstream retrieval consistency on messy conversational inputs.
- Fine-tuned transformer models and integrated LLM-driven structured filter extraction (numerical and categorical constraints), combining dense retrieval with rule-aware post-filtering to significantly improve Precision@K and Recall@K metrics.
- Collaborated cross-functionally with product and platform teams to productionize GenAI systems, optimizing latency, throughput, and cost, and delivering measurable improvements in search relevance, user satisfaction, and downstream conversion signals.

Client: Hitachi GlobalLogic Hyderabad, India

Nov 2021 – Dec 2022

Senior ML Engineer

- Designed and deployed deep learning pipelines using TensorFlow/Keras and attention based LSTM architectures for demand forecasting across 100,000+ SKUs, achieving ~96% accuracy on hold out test sets and reducing forecast error by ~35% versus legacy methods.
- Engineered the full MLOps lifecycle on AWS SageMaker—including model training, hyperparameter tuning, batch endpoints and real time inference endpoints—serving features and predictions for 10+ production models, and reduced model deployment latency by ~40% through automation and standardized deployment workflows.
- Architected data ingestion, cleansing and feature engineering workflows in MySQL, Hive and Apache Airflow; leveraged Databricks for scalable collaborative data processing, enabling handling of terabytes of data and supporting cross team analytics and training pipelines.
- Built a real time anomaly detection pipeline using Python, Kafka streaming and alerting logic to monitor order placement drops, app responsiveness and fulfillment bottlenecks—enabling proactive incident detection and reducing mean time to resolution for customer impacting issues by ~30%.
- Applied NLP techniques on product titles/descriptions and customer interaction text (search queries, feedback, support tags) to improve product taxonomy quality and downstream personalization signals, strengthening feature sets used in forecasting/recommendation workflows.
- Partnered with marketing, supply chain, operations and inventory teams to translate business questions into ML problems: e.g., campaign uplift modelling, supply chain repositioning, SKU level inventory optimization; generated dashboards and insights to embed ML output into business decision loops.
- Conducted rigorous A/B testing, multivariate experimentation and user segmentation analysis to support growth and conversion strategies; provided statistical insights and model driven recommendations used to prioritise product roadmap items and optimise user journeys.
- Led model versioning, monitoring and drift detection frameworks: implemented model performance dashboards, automated retraining triggers, rollback strategies, and shadow traffic experiments to maintain model fidelity in production under evolving datasets and seasonal shifts.
- Developed and integrated CI/CD pipelines using Jenkins, GitLab CI, Docker and Kubernetes for ML artifact packaging and deployment across staging and production environments; enforced code reviews, unit/integration tests, data contracts and reproducible pipelines to ensure robust and auditable deployments.
- Mentored and collaborated with data scientists and ML engineers on best practices (feature engineering, model interpretability, scalability, latency optimisation) and worked closely with infra/devops teams to ensure production readiness, cost efficiency (e.g., spot instances, model quantisation) and operational robustness.
- Served as a key technical liaison and stakeholder partner: defined ML strategy aligned to Hitachi's e commerce scale requirements (e.g., demand forecasting, dynamic pricing, personalization), evaluated trade offs between time to market vs model complexity, and advised leadership on AI/ML tool stack adoption and data governance practices.

Environment: Python, TensorFlow, AWS SageMaker, Databricks, MySQL, Hive, Apache Airflow, Kafka, Jenkins/GitLab CI, Docker, Kubernetes, SQL, visualization & dashboarding tools. Production monitoring

Client: Lytx Chennai, India

Nov 2018 – Oct 2021

Data Scientist

- Led a globally distributed analytics & DevOps team spanning the US, Argentina and Poland, driving delivery of big data solutions to process and analyze millions of telematics events per day for fleet safety, vehicle usage and

operational efficiency.

- Architected and built high throughput SQL and Databricks Delta Lake pipelines ingesting 6M+ daily telematics events (video, sensor, GPS, vehicle logs), ensuring >99.9% data freshness and high quality availability for real time analytics and modelling of driver behaviour, risk events, and fleet performance.
- Established semantic layer via dbt Core, defined reusable data marts and standardized metrics for self service analytics across product, safety, operations and compliance functions — enabling advanced cohort and A/B test analytics to measure behavioural change and fleet risk reduction.
- Designed and deployed automated workflows using AWS Glue and Apache Airflow to orchestrate ingestion and transformation of campaign, CRM, compliance and fleet telematics datasets; developed interactive dashboards (e.g., Tableau) reducing reporting turnaround time by 40% and empowering decision makers with near real time insights.
- Built and maintained dimensional data models (star and snowflake schemas) in platforms such as Snowflake and Databricks to support scalable analytics across sales, operations, risk & compliance teams — delivering flexible and performant ad hoc reporting and large scale cohort segmentation.
- Provided critical production support and performance tuning for global deployments — diagnosing latency issues, scaling Spark/Databricks clusters, optimizing SQL/Hive queries, and collaborating with site reliability/DevOps teams to maintain high data pipeline reliability, incident response readiness, and availability in a mission critical telematics environment.
- Partnered closely with product, safety, operations and behavioural science stakeholders to design and analyse 10+ A/B tests (e.g., driver coaching interventions, risk alert triggers), translating raw results into actionable insights that improved safety conversion metrics and drove measurable business impact (e.g., reduced risky event rate by ~8%).
- Mentored junior data scientists and data engineers, promoted best practice modelling (feature engineering for telematics sensors, anomaly detection, behavioural scoring), shepherded adoption of version control, reproducible workflows, and runbook-style documentation for recurring operational issues.

Environment: Python, SQL/MySQL, Snowflake, Databricks with Delta Lake, dbt Core, AWS Glue, Apache Airflow, Tableau, Kafka (real-time stream processing), Docker/Kubernetes (for pipelines & deployments), Spark (in Databricks), Git, Jenkins/GitLab CI, REST/JSON APIs, version-control, orchestration frameworks, cloud storage & compute.

Client: Geetham Software Chennai, India

Jul 2015 – Oct 2018

Data Scientist

- Built and maintained ETL pipelines across the bank's Azure based data ecosystem for marketing campaigns, finance-ledger data and CRM systems; introduced standardized data quality checks, metadata lineage tracking and SLA visibility dashboards to support audit ready business intelligence.
- Developed segmentation, churn prediction and propensity scoring models that fed inbound and outbound CRM campaigns; collaborated with business & analytics teams to extract quarterly business review KPIs and summarize insights for executive leadership.
- Owned operational reliability of data pipelines—implemented monitoring, automated failure handling logic, root cause analysis (RCA) routines, and worked upstream with source system stakeholders to establish formal data contracts reducing recurring breaks by ~30%.
- Established data governance protocols across naming conventions, PII / sensitive data handling, access roles, encryption at rest and in flight, and data retention policies; partnered with Legal, Compliance and Finance functions to codify approved usage of sensitive fields and support regulatory audit readiness.
- Collaborated with product managers and CRM leads to translate insights into testable business hypotheses, implemented hold out and cohort analyses of campaign/offer effectiveness, and measured uplift and ROI for major customer journey initiatives.
- Instituted self service analytic frameworks via semantic data models and reusable data marts (built in Azure Data Lake / Synapse) enabling business analysts to generate insights without heavy engineering involvement — reducing time to insight by ~40%.
- Directed cross functional analytics engagements: worked with marketing, risk, finance, operations and channel management teams to embed model outputs into live business processes (customer retention triggers, next best offer scoring, campaign optimisation) driving measurable uplift in conversion and retention.

- Mentored junior data scientists and data engineers in best practice feature engineering, modelling and documentation; set up version control/reproducibility standards, and defined the strategic analytics roadmap aligned with the bank's broader digital transformation in the banking & financial services sector.

Environment: Python, SQL / Azure SQL, Azure Data Lake, Azure Synapse Analytics, dbt, Airflow, Azure Data Factory, Power BI (or equivalent), Git, Jenkins (or Azure DevOps CI), REST/JSON APIs, Azure Active Directory (RBAC), encryption-at-rest/in-flight, naming/data-governance frameworks.

Education:

- Master of Science in Data Science, The University of Arizona Tucson, AZ 2024
- Bachelor of Technology in Mechanical Engineering, Acharya Nagarjuna University Guntur, India 2015

Research Paper:

LLM-based Semantic Search for Conversational Queries in E-commerce

- Developed a scalable semantic-search engine for e-commerce by fine-tuning Sentence Transformers and Google FLAN-T5 on a 1.3 M-product catalog, integrating FAISS for low-latency vector retrieval.
- Led applied research culminating in a peer-reviewed paper ("LLM-based Semantic Search for Conversational Queries in E-commerce") submitted to CIKM 2025, demonstrating a 90% boost in Precision@10 over traditional keyword search.

Projects:

Building LLM From Scratch.

- Implemented GPT-style model (PyTorch, byte-level tokenizer, scaled dot-product attention) with full training + SFT pipeline; documented acceptance tests, perf baselines, and rollback checkpoints for experiments.
- Drove R&D prototyping and ablation studies (stability, context handling) to inform architecture decisions; created a tech decision log and "go/no-go" criteria for releases.

Predicting Customer Churn in E-Commerce

- Built churn models (Random Forest/SVM/XGBoost, 500K+ records) with >90% accuracy; delivered Tableau dashboards for cohort retention and SLAs/KPIs tied to product goals.
- Shipped **reproducible pipelines** (CV/grid search, data quality checks) with a **UAT checklist and runbook**, mirroring audit-ready analytics in regulated environments.

Enterprise Knowledge Copilot: Agentic RAG for Internal Q&A & SOP Automation

- Built a LangChain + pgVector/Redis RAG system to query SOPs/runbooks/policies; modeled change-management docs and created access controls for sensitive content.
- Produced release notes, incident triage flows, and telemetry on question latency/answer confidence, demonstrating integration governance and operability.

High-Performance Video Diffusion Model Optimization – Wan 2.2

- Optimized the Wan 2.2 text-to-video diffusion model pipeline for H100 GPUs, leveraging advanced context-parallelism, flash attention 3, and direct cuBLAS/triton kernel integration to achieve sub-25s 720p video generation for 81 frames—outperforming standard FA2/FA3 baselines (15–30 TFLOPS, 2–4× speedup).

Certifications:

- Deep Learning Specialization – Andrew Ng, Coursera
- SQL(Advanced) - Hackerrank